

House Price Prediction - Project Summary

Objective

The main goal of this project was to build a machine learning model that can predict house prices based on different property features. By analyzing the dataset, I wanted to understand which factors have the biggest impact on house prices and compare the performance of different regression models.

Dataset Description

The dataset used for this project was the Housing Prices Dataset from Kaggle. It contains information about houses such as area, number of bedrooms, bathrooms, parking spaces, furnishing status, air conditioning, preferred area, and other features. During the data exploration process, I found that the dataset had no missing values and no duplicate records, which made the data cleaning process simple.

Data Preparation and Model Building

To prepare the data for machine learning, categorical columns such as furnishing status, air conditioning, and preferred area were converted into numerical values using one-hot encoding. The dataset was then split into training and testing sets in an 80:20 ratio. Two models were trained: Linear Regression and Random Forest Regressor.

After evaluating both models, Linear Regression performed slightly better. It achieved an MAE of 970,043, an RMSE of 1,324,507, and an R^2 score of 0.65. The Random Forest model achieved an R^2 score of 0.61. Therefore, Linear Regression was selected as the better model for this dataset.

Analysis and Visualizations

Several visualizations were created to better understand the data.

- The house price distribution histogram showed that most houses were concentrated in the middle price range, while only a few houses had very high prices.
- The correlation heatmap helped identify the features most related to house prices, with area, bathrooms, air conditioning, parking, and preferred area showing stronger positive relationships.
- The furnishing status boxplot revealed that furnished houses generally had higher prices than semi-furnished and unfurnished houses.

- Finally, the Actual vs Predicted Price scatter plot showed a clear positive trend, indicating that the model was able to capture the overall relationship between property features and house prices.

Conclusion

Overall, the project was successful in predicting house prices with reasonable accuracy. The Linear Regression model was able to explain about 65% of the variation in house prices, which indicates a fairly good performance. One interesting observation was that amenities such as air conditioning and furnishing status had a noticeable impact on property value. Based on these findings, real estate businesses should focus on highlighting property size, furnishing quality, parking facilities, and other amenities when marketing houses, as these features are strongly related to higher selling prices.